This means, **for a bar magnet the intensity of magnetisation can be defined as the pole strength per unit area (face area).**

(d) Magnetic induction or total magnetic field

When a substance like soft iron bar is placed in a uniform magnetising field $\vec{H}$, the substance gets magnetised. **The magnetic induction (total magnetic field) inside the specimen $\vec{B}$ is equal to the sum of the magnetic field $\vec{B}_o$ produced in vacuum due to the magnetising field and the magnetic field $\vec{B}_m$ due to the induced magnetism of the substance.**

$$\vec{B} = \vec{B}_o + \vec{B}_m = \mu_\circ \vec{H} + \mu_\circ \vec{M}$$

$$\Rightarrow \vec{B} = \vec{B}_o + \vec{B}_m = \mu_\circ (\vec{H} + \vec{M}) \qquad (3.32)$$

(e) Magnetic susceptibility

When a substance is kept in a magnetising field $\vec{H}$, magnetic susceptibility gives information about how a material responds to the external (applied) magnetic field. In other words, the magnetic susceptibility measures how easily and how strongly a material can be magnetised. It is defined **as the ratio of the intensity of magnetisation $(\vec{M})$ induced in the material to the magnetising field $(\vec{H})$**

$$\chi_m = \frac{|\vec{M}|}{|\vec{H}|} \qquad (3.33)$$

It is a dimensionless quantity. Magnetic susceptibility for some of the isotropic substances is given in Table 3.1.

Table 3.1 Magnetic susceptibility for various materials

Material	Magnetic susceptibility (χ_m)
Aluminium	2.3×10^{-5}
Copper	-0.98×10^{-5}
Diamond	-2.2×10^{-5}
Gold	-3.6×10^{-5}
Mercury	-3.2×10^{-5}
Silver	-2.6×10^{-5}
Titanium	7.06×10^{-5}
Tungsten	6.8×10^{-5}
Carbon dioxide (1 atm)	-2.3×10^{-9}
Oxygen (1 atm)	2090×10^{-9}

EXAMPLE 3.8

Compute the intensity of magnetisation of the bar magnet whose mass, magnetic moment and density are 200 g, 2 A m^2 and 8 g cm^{-3}, respectively.

Solution

Density of the magnet is

$$Density = \frac{Mass}{Volume} \Rightarrow Volume = \frac{Mass}{Density}$$

$$Volume = \frac{200 \times 10^{-3}\,\text{kg}}{\left(8 \times 10^{-3}\,\text{kg}\right) \times 10^{6}\,\text{m}^{-3}}$$

$$= 25 \times 10^{-6}\,\text{m}^3$$

Magnitude of magnetic moment $p_m = 2\,\text{A m}^2$

Intensity of magnetization,

$$M = \frac{Magnetic\,moment}{Volume} = \frac{2}{25 \times 10^{-6}}$$

$$M = 0.8 \times 10^{5}\,\text{A m}^{-1}$$

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